

National University of Computer and Emerging Sciences (FAST)

Applied Calculus(EE)

Mid-I (Fall' 2015)

Roll No: 152-4522

Section: EE-2

Time allowed: 60min

Max.Marks:40

Please read the following instructions carefully before attempting any of the questions:

1. Attempt all questions, all questions carry equal marks.
2. Do not ask any question about the contents of this examination from anyone.
 - a. If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.
 - b. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.
 - c. Write all steps, missing steps may lead to deduction of marks.

Q1. Solve the following inequality

$$\frac{x-2}{x+3} \leq 1$$

Q2. Find the following limit or explain why the limit does not exist?

$$\lim_{x \rightarrow -1} \frac{2}{(x+1)} |x+1|$$

Is the function continuous at $x=-1$?

Q3. Evaluate $\lim_{x \rightarrow 2} \frac{1}{(x-2)}$

Also find asymptotes for the curve.

Q4. Graph the function, not by plotting points, but starting with the graph of one of the standard functions and applying the appropriate transformations.

$$y = 1 - 2\sqrt{x+3}$$

National University of Computer and Emerging Sciences

Semester: Fall 2015

Course: Applied Calculus (EE) Mid: 2

Time: 1 hr

Total Marks: 30

Question.1: Find the critical points of

$$f(x) = x^{1/3}(x - 4)$$

Identify the intervals on which f is increasing and decreasing. Find the function's local and absolute extreme values.

Question.2: Evaluate the following integral, if exist and give reason if does not exist.

$$\int_{-1}^2 (x - 2|x|)dx$$

Question.3: Sketch the region enclosed by the given curves. Describe whether to integrate with respect to x or y (with reason). Then find the area of the region.

$$y = 8 - x^2, \quad y = x^2, \quad x = -3, \quad x = 3$$

National University of Computer and Emerging Sciences (FAST)

EE

Applied Calculus (EE) Final (Fall 2015)

Time allowed: 3 hours

Max.Marks:100

Please read the following instructions carefully before attempting any of the questions:

- Attempt all questions, all questions carry equal marks.
- Do not ask any question about the contents of this examination from anyone.
- If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.
- If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.
- Write all steps, missing steps may lead to deduction of marks.

Question.1: Find the following limits

(a) $\lim_{x \rightarrow 1} \frac{x-1}{|x-1|}$

(b) $\lim_{x \rightarrow \infty} \frac{\sin 2x}{x}$

(c) Define $f(1)$ in a way that extends $f(x) = \frac{x^3-1}{x^2-1}$ to be continuous at $x=1$.

Question.2: Sketch a graph of the function $g(x) = \frac{(x+1)^2}{1+x^2}$ using the following steps.

- Identify where the extrema of g occur.
- Find the intervals on which g is increasing and the intervals on which g is decreasing.
- Find where the graph of g is concave up and where it is concave down.
- Sketch the general shape of the graph for g .

Question.3:

- Find the area of the regions enclosed by the curves $x - y^2 = 0$ and $x + 2y^2 = 3$.
- Evaluate $\iint_R y \sin(xy) dA$, where $R = [1, 2] \times [0, \pi]$.

Question.4:

- Solve the following improper integral

$$\int_0^1 x \ln x \, dx$$

- Is the following integral convergent or divergent? Give reason for your answer. (Use comparison theorem)

$$\int_1^{\infty} \frac{e^x}{x} \, dx$$

Question.5:

- Find a plane through $P_1(1, 2, 3)$, $P_2(3, 2, 1)$ and perpendicular to the plane $4x - y + 2z = 7$.
- Solve the inequality, express the solution set as interval or union of intervals. Also, show the solution on the real line.

$$\left| \frac{2}{x} - 4 \right| < 3$$